

EMPLOYEE ATTRITION PREDICTION

Summary

We analyzed data from 1,470 employees to understand why people leave — and, more importantly, to identify who is most at risk of leaving next. Using three different predictive models, we found that employee departures are not random: they follow clear, consistent patterns that HR can act on right now.

Our best-performing model correctly identifies at-risk employees well above chance, giving the HR team a practical early-warning tool to intervene before a resignation letter lands on a desk.

What the Data Tells Us

Current State of Attrition

- 16% of employees left the company.
- The Sales department has the highest exit rate at 27.3%
- Human Resources roles and Laboratory Technicians are the most at-risk job types
- Employees who left earned a median salary of \$3,202/month vs \$5,204/month for those who stayed

The Top 3 Reasons Employees Leave

After running our analysis, three factors stood out above everything else as the strongest signals that an employee is likely to leave:

#	Factor	What It Means
1	Job Role (HR Department)	Employees in Human Resources roles show the highest departure rates, suggesting workload, pay, or recognition gaps specific to this function.
2	Distance from Home	Longer commutes significantly increase the likelihood of leaving. Employees traveling more than 20 km are notably higher risk.
3	Years Since Last Promotion	Employees who have not had a promotion or career progression in several years feel stagnant and are far more likely to seek opportunities elsewhere.

Is It Just About Pay?

No — and this is one of the most important findings for HR leadership. While employees who left did earn less on average (\$3,857 vs \$5,118/month), salary ranked below factors like job satisfaction, commute distance, and career progression in our model.

This means a pay raise alone will not stop attrition. Employees are leaving because they feel stuck in their careers, burned out from commuting or overwork, and disengaged from their day-to-day role. A targeted approach is needed — not a blanket salary increase.

2 Recommendations HR Can Act On Today

Recommendation 1: Launch a 18-Month Career Review Cycle

Every employee who has not received a promotion, role change, or formal recognition within the past 18 months should be flagged for a mandatory career conversation with their manager. Our data shows that stagnation is one of the top three exit triggers. Even a lateral growth opportunity — a new project, a team lead responsibility, or skills development investment — reduces the psychological itch to leave. Start with Human Resources and Sales, where attrition is highest.

Recommendation 2: Pilot a Commute Relief Programme

Employees commuting more than 20 km are disproportionately represented among those who leave. HR should introduce flexible work arrangements (hybrid/remote 2 days/week) specifically for this group. Alternatively, a commute subsidy or transport allowance for high-distance employees costs far less than replacing them. The average cost of replacing one employee is estimated at 50–70% of their annual salary — a commute perk is a fraction of that.

One Important Caution

This model is a prediction tool, not a verdict. It identifies patterns from historical data and flags risk — it does not guarantee that a flagged employee will leave, nor does it mean an unflagged employee is safe. HR should treat model outputs as a starting point for a conversation, not a conclusion.

The model should be updated every 6–12 months to remain relevant as the organisation evolves. Used responsibly, it gives HR a significant head start in retention.

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